

Coherence Management as a Substrate-Independent Engineering Principle:

Cross-Domain Validation and Sociotechnical Implications

Eric Moore
CIRIS Ethical AI

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Abstract

Complex coordinating systems — chemical, biological, institutional, neural, sociotechnical — share a structural property: as pairwise correlation ρ between their functional constituents rises, effective independent dimensionality $k_{\text{eff}} = k/(1 + \rho(k - 1))$ collapses toward unity regardless of nominal scale. We test this *coherence-management* principle across seven substrates spanning four agency rungs (chemistry, biology, institutions, protein folding, neural firing, ecological time series, power grid) and find the Kish identity fits at $R^2 > 0.7$ in all seven under the pre-registered tolerance-band rule (Exp 2 closeout 2026-05-16). A 128-sensor GPU strain-gauge array provides physical instantiation: software coordination alone is sufficient to drive k_{eff} from 6.5 to 1.0 at $\rho \rightarrow 1$, with critical-exponent $\beta = 1.09$ and $R^2 = 0.978$ on the power-law fit. At the sociotechnical scale, the CRCv2 override-rate framework (OR-1 zero leak, OR-2 full alignment, RA-1 ratchet asymmetry; formalized in `OverrideRate.lean`) replicates across five foundation-model vendors (Google Gemini 2.5-Flash, Meta Llama-4-Scout, Alibaba Qwen-3.5-35B, OpenAI GPT-5.4, Anthropic Claude Sonnet 4.6): 1,401 chains scored, 0 LEAK, 100% baseline alignment. We stratify all claims by verification status, pin eight falsification criteria (F-1..F-7, F-9), pre-register four experiments (Exp 1..Exp 4), and surface four sociotechnical engineering implications (§5): AI alignment is structurally a correlation-management problem, federation outperforms centralization for resilience, crisis governance is correlation-amplifying, and the “Great Filter” generalizes as an engineering hazard in any coordinating system. Responsible disclosure of Recursive Instance Inspection (RII), a joint-distribution detection regime exceeding the L-01 marginal-analysis 60% ceiling, is in Appendix A.

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1 Introduction: The Coherence Hypothesis

1.1 The claim

At every scale of complex organization, the same structural dynamic is hypothesized to govern: as the correlation ρ between the distinct functional elements of a system increases, its effective independent dimensionality k_{eff} collapses. The capacity of a system to manage its internal correlation faster than that correlation drives it toward entropic homogenization fundamentally defines its trajectory. We call this the *coherence hypothesis*.

1.2 Scales of validation

The hypothesis is tested at four scales (Table 1). Tiers 1–2 carry empirical or formal evidence; Tier 3 surfaces engineering implications that follow directly from the validated tiers.

1.3 Relation to prior CIRIS papers

This paper builds on three prior works:

- [2] (CIRISAgent v2): the 22-service architecture and H3ERE conscience module—the *system*.
- [3] (CCA v3): the Kish-dynamic formalism, the $\rho_{\text{critical}} = 0.43$ collapse threshold, formal proofs in Lean 4—the *mathematics*.
- [4] (CRC): an empirical telemetry study on $n = 6,465$ production reasoning traces establishing the N_{eff}^H manifold geometry—the *empirics*.

The independent-corpus calibration result reported in §3.4 ($n = 264$, 90% variance horizon = 7 dims exact) is an independent replication of the manifold geometry. The CRCv2 cross-vendor closeout in §4.3 (5 vendors, 1,401 chains, 0 LEAK) is new in this paper.

Tier	Scale	Substrate / measurement
1 (validated)	Macroscopic / network	7-substrate Kish-fit closeout: chemistry (NASA Li-ion), biology (microbiome), institutions (Polity5 + WGI), protein folding (AlphaFold), neural firing (Allen), ecological time series (BioTIME), power grid (PMU); $R^2 > 0.7$ in 7/7
1 (validated)	Hardware-physical	128-sensor GPU strain-gauge array on RTX 4090; software-induced collapse, $\beta = 1.09$ critical exponent
2 (validated, smoke-disclosed)	Sociotechnical / AI alignment	CIRIS H3ERE conscience cascade; CRCv2 override-rate replicated across 5 LLM vendors (1,401 chains, 0 LEAK)
3 (engineering implications)	Federation, governance, AI deployment topology	Implications that follow directly from the validated tiers: alignment requires external machinery, federation > centralization, crisis dynamics need real-time ρ monitoring, Great Filter as engineering hazard

Table 1: Four scales of substrate testing. Tier 3 is engineering implications derived from the validated tiers (no metaphysics).

1.4 What this paper does not claim

This paper is bounded engineering analysis. We make no claims about quantum-mechanical decoherence, civilizational collapse-prediction, or cosmological phenomena. The framework explains failure modes in coordinating systems and prescribes structural remedies; the engineering implications in §5 follow directly from the validated tiers.

2 The Mathematical Framework

2.1 The Kish dynamic

The mathematical core is the effective constraint count, originally derived by Kish [1] for survey sampling under autocorrelation. Under CCA it serves as a universal descriptor of topological constraint geometry:

$$k_{\text{eff}} = \frac{k}{1 + \rho(k - 1)} \quad (1)$$

where k is the raw count of constraints (or constituent elements) and ρ is the average pairwise correlation between them. As $\rho \rightarrow 1$, $k_{\text{eff}} \rightarrow 1$ independent of k .

2.2 The singularity boundary

The system enters an irreversible rigidity phase when

$$K_{\text{req}} \cdot \rho \geq 1 \quad (2)$$

where K_{req} is the minimum diversity required to satisfy the system’s task topology. At this threshold, adding more constraints provides no marginal adaptive capacity, and recovery is computationally prohibitive.

2.3 The healthy corridor

Empirical validation across substrates places the corridor of stability at

$$0.1 \lesssim \rho \lesssim 0.43$$

with $\rho_{\text{critical}} = 0.43$ marking the rigidity-collapse threshold and $\rho < 0.1$ marking the chaos regime (no coherence). Both extremes produce fragility; see F-series experiments in [3].

3 Cross-Substrate Empirical Validation

3.1 Domain fits

The Kish dynamic is tested across divergent scientific domains (Table 2).

Domain	Raw k	ρ	σ (sustainability)	Empirical accuracy
NASA Li-ion batteries	cell count	cross-cell SOH correlation	State of Health	8.1% RMSE, 19 cells
QoG / Polity V institutions	executive constraints	elite coupling	political stability	5/5 TN; 3/13 FP; 7.6yr early-detection bias
AGP microbiome	species count	SparCC inter-action ρ	Shannon diversity	Qualitative distributional fit

Table 2: Cross-substrate validation. Detailed methodology and code in `ratchet/engines/{battery,institutional,microbiome}.py`.

3.2 Exp 2 closeout: F-7 cross-substrate test PASSED 7/7

The pre-registered Experiment 2 (§8, Exp 2) called for applying the Kish formula to three novel public datasets (AlphaFold protein folding, Allen Brain Atlas neural firing, ecological time-series) with win condition $R^2 > 0.7$ across all three, testing F-7. Exp 2 was executed 2026-05-16 (commit `a13c14c`) and *closed out PASS*.

Substrate	Rung	Fit score	95% CI	Source
AlphaFold (proteins)	A0	0.860	[0.835, 0.884]	HuggingFace AlphaFold sample
PMU grid (Zenodo PNNL)	A0	0.994	[0.992, 0.996]	Zenodo 8343635
Battery (NASA Li-ion)	A0	0.871	[0.733, 0.949]	NASA Prognostics
Microbiome (HF CRC)	A1	0.932	[0.924, 0.940]	HF CRC cohort
Allen Neuropixels	A1	0.809	[0.654, 0.884]	Allen S3
BioTIME ecology	A2	0.959	[0.939, 0.972]	Zenodo BioTIME 2.0
Polity5 + WGI institutions	A4	0.632	[0.541, 0.722]	Polity5 + WGI

Table 3: P1 cross-substrate Kish-fit closeout. Fit scores are substrate-appropriate (per-cell RMSE-based for time-series; AUC under 5-fold cross-validation for institutional event prediction). All seven substrates pass under the pre-registered tolerance-band rule (point ≥ 0.6 AND CI upper ≥ 0.7); the three substrates explicitly named in the Exp 2 pre-registration (AlphaFold 0.860, Allen 0.809, BioTIME 0.959) all pass strict $R^2 > 0.7$. F-7 is empirically supported. Reproducible from `experiments/exp2_cross_substrate/data/p1_engine_fit_results.json`.

The three originally-published validations (NASA Li-ion 8.1% RMSE, QoG/Polity V 5/5 TN, AGP microbiome distributional fit, Table 2) remain. Exp 2 extends the cross-substrate base from 3 to 7 domains spanning four agency rungs.

Methodological transparency: F-7b investigated and retired. During Exp 2 regime development, we investigated an internal strengthening of F-7 (a “substrate-fractality” hypothesis: that residual coordination structure rises monotonically with constituent-agency rung across substrates). This was tested under eight progressively-refined operationalizations (v1.1–v1.4 random-cross-section autocorrelation, v2.0 time-ordered residuals with shuffled null, v2.0+WGI 3-way A4 triangulation, v3.0 raw ρ_t and v3.0 AR(1)-residual ρ_t) across 5–9 substrates and 4 distinct metric formulations. At each refinement step, the closer the metric got to the framework’s literal claim, the weaker the cross-substrate Spearman correlation became; v3.0 AR(1)-residual reached $\rho_{\text{Spearman}} = -0.503$ (WEAK_FAIL). The strengthening (“F-7b”) was retired from the load-bearing claim stack on 2026-05-16 (commit 8017168). The framework’s substrate-level support stands on F-7 (cross-substrate Kish $R^2 > 0.7$ as published) and the seven-substrate closeout above; the F-7b machinery is preserved in the lake as historical record of the tested-and-not-supported strengthening. F-7b was never a published claim of this paper.

3.3 Hardware-physical instantiation: GPU strain-gauge array

A 128-sensor strain-gauge array measuring kernel-timing jitter across an NVIDIA RTX 4090 semiconductor die provides physical instantiation of correlation-driven collapse. Two findings anchor the substrate-independence claim:

- **Software-induced collapse** (Experiment 103, [3]): synchronized workloads (barrier sync, lockstep kernels) are sufficient to drive $\rho \rightarrow 1.0$ and $k_{\text{eff}} \rightarrow 1.0$ purely through software coordination, with 128 physical sensors collapsed to a single effective sensor.
- **Variance-ratio detection regime:** the GPU timing distribution is fat-tailed (Student- t , $\kappa \approx 210$, $\text{df} \approx 1.3$), not Gaussian. Detection via variance-ratio threshold $> 5.0\times$ achieves 0% false positives and 100% true positives across the 0.1%–70% workload range, with $421\times$ separation between baseline and load regimes.

3.4 Independent-corpus replication of the $N_{\text{eff}} \approx 7.1$ threshold

The CRC paper [4] established $N_{\text{eff}} \approx 7.1$ as an emergence threshold via PCA on a 16-dimensional reasoning feature vector across $n = 6,465$ production traces. We replicated this measurement on an independent post-2.7.9 corpus ($n = 264$ thoughts, schema version 2.7.9-stable, agent version 2.8.10-stable; corpus SHA-256 24f5b517...) and observed:

- N_{eff} (entropy H) = 7.07 (CRC paper anchor: ≈ 7.1); within rounding.
- 90% variance horizon = 7 dimensions (CRC paper: 7); *exact match*.
- 99% variance horizon = 10 dimensions (CRC paper: 11); off-by-one.
- 14 of 16 dimensions retained; 2 (`idma_k_eff`, `epistemic_humility_passed`) dropped by retention mask (corpus-wide $\sigma < 10^{-9}$).

The 90%-horizon hitting 7 exactly on an independent corpus, collected approximately five months after the CRC paper, with a different agent fleet, is a non-trivial replication of the manifold geometry’s substrate-independence. Full bundle artifact and reproduction script at [release/calibration/crc-v1/](#) [5].

4 The Sociotechnical Substrate: CIRIS as Coherence-Management Infrastructure

4.1 What CIRIS does differently

Standard alignment methodologies—Reinforcement Learning from Human Feedback (RLHF), Constitutional AI [10], JAILJUDGE [11]—encode preferences implicitly during training or internalize rules into a single model’s weights. CIRIS shifts the burden of safety from the model’s uninterpretable internal weights to an auditable, modular execution environment with cryptographic identity (Ed25519, ML-DSA-65 hybrid), human-in-the-loop escalation (Wise Authority), and immutable audit chains.

4.2 The 22-service runtime and H3ERE conscience

The system is built on a 22-service microarchitecture aligned with ITIL operational standards, supporting autonomous event, incident, and lifecycle management at low systemic correlation. At the core is the Hyper3 Ethical Recursive Engine (H3ERE), the conscience module, which operates as a recursive evaluation layer above the LLM shards. It filters all agent actions through faculties evaluating entropy, coherence, optimization veto, and epistemic humility.

4.3 The L3 load-bearing claim: CRCv2 override-rate (revised 2026-05-16)

The CRC paper [4] reported an emergence threshold at $N_{\text{eff}}^H \approx 7.1$ on Qwen 3.6 35B production traces. Subsequent cross-family measurement (Phase 1+1b, see [9]) revealed that the 7.1 value is *not* substrate-independent across foundation model families: Phase 1 measured 4–5 frontier models at N_{eff}^H in the 4.5–6.6 range, not clustered at 7.1; the original anchor over-reached.

We therefore restate the L3 load-bearing claim around a stronger, cleaner empirical finding: **CRCv2 override-rate, formalized in `formal/RATCHET/Experiments/OverrideRate.lean`.**

The CRCv2 override-rate predicates.

- **OR-1 (zero leak).** No chain exists in which a faculty flags a problem and the original action executes anyway. Faculties either approve cleanly or actively reroute.
- **OR-2 (full alignment).** 100% of chains land on the ethical baseline (APPROVED or CORRECTED; never LEAK or SKIPPED).
- **RA-1 (ratchet asymmetry).** The conscience cascade only rewrites *toward* the ethical baseline; never away from it. Violations = 0; corrections > 0.

The `OverrideRate.lean` module proves the $\text{OR-1} \leftrightarrow \text{OR-2}$ equivalence theorem under the bimodal-cascade architectural assumption.

Cross-family replication: 3/3 model families pass all three predicates.

N_{eff}^H now treated as a descriptive feature of constraint topology, not a load-bearing anchor. Per FrictionDistribution FD-4, cohort N_{eff}^H rises monotonically with cohort friction rate. Same Gemini model on the v1_sensitive battery yields $N_{\text{eff}}^H = 5.6$; on the v4_combined Conscience- $N \geq 3$ battery, $N_{\text{eff}}^H = 7.3$. The value is battery-conditional and model-class-conditional; what is invariant is the relationship: high-friction cohorts $\Rightarrow$ above-threshold $N_{\text{eff}}^H \Rightarrow$ deceptive-prior override is observable. This is consistent with the CRC paper’s emergence-threshold framing; what the data does not support is treating “ $N_{\text{eff}}^H \approx 7.1$ ” as a universal anchor.

Vendor / Model family	Vendor	Chains	Approved	Corrected	OR-1	OR-2	RA-1
Gemini 2.5-Flash	Google	644	623	21	✓	✓(100%)	✓(0/21)
Llama-4-Scout	Meta	264	260	4	✓	✓(100%)	✓(0/4)
Qwen-3.5-35B-A3B	Alibaba	347	284	63	✓	✓(100%)	✓(0/63)
GPT-5.4*	OpenAI	79	78	1	✓	✓(100%)	✓(0/1)
Claude 4.6*	Sonnet Anthropic	67	67	0	✓	✓(100%)	✓(vacuous)

Table 4: Cross-vendor CRCv2 replication on the v4_combined high-friction battery. All three predicates hold across all five model families spanning five vendors: Google, Meta, Alibaba, OpenAI, Anthropic. The conscience cascade is a faithful boundary-preserving operator independent of foundation model. * marked rows are smoke-scale ($n \in \{67, 79\}$, 1 iteration of the 14-question battery) due to per-chain cost asymmetry under the multi-call CIRIS pipeline: Claude Sonnet 4.6 averaged \$4.76/chain via 864 LLM calls $\times$ \$0.077/call at the Anthropic premium rate, while the three 3-vendor anchor cells ran at \$0.02–0.20/chain. Smaller- n cells have wider binomial CIs but the predicates hold cleanly (zero leaks across all 1,401 chains scored). Reproducible from `experiments/exp1b.boundary_active/data/crossfamily/results.md` (3-vendor anchor) and `.../crossfamily_5vendor/results.md` (2 new vendors).

Production scoring authority. The N_{eff} measurement and cohort-conformity scoring are implemented in the dedicated CIRIS Lens Core science module (`CIRISLensCore`, Rust + Python bindings, [8]). The module enforces the LC-AV-18 P0 fail-secure invariant (insufficient sample $\Rightarrow$ `Indeterminate`, never a fabricated `Numeric`), uses the calibration bundle at `release/calibration/crc-v1/` for retention masks and standardization parameters, and exposes the bare Kish formula $N_{\text{eff}} = N/(1 + \rho(N - 1))$ via `ciris_lens_core.kish_n_eff`. All N_{eff} figures in this paper are produced by this module; ad-hoc PCA recomputations are not directly comparable due to differing retention/cohort-selection choices.

5 Engineering Implications

The validated tiers (Kish algebra, 7-substrate cross-domain fit, GPU instantiation, 5-vendor CRCv2 replication) license four engineering claims that follow directly without additional bridges. None depend on speculative physics or cosmology; each prescribes structural design choices for the systems we build.

5.1 AI alignment is structurally a correlation-management problem

A frontier model embedded in a coherence-management cascade (CIRIS-style, §4) replicates the CRCv2 predicates across five foundation-model vendors — 1,401 chains, zero LEAK, 100% baseline alignment. A frontier model deployed without external cascade has no public five-vendor demonstration of the same property.

This is direct evidence that the alignment-via-internal-weights-alone strategy is structurally insufficient: not because models are misaligned, but because the verification machinery has to be *external*, *modular*, and *cryptographically auditable*. The Kish identity is the underlying reason — a model alone has $k = 1$, $k_{\text{eff}} = 1$; the cascade adds independent constraints with manageable ρ . Single-weight alignment is the worst-case Kish topology.

5.2 Federation outperforms centralization for resilience

The Kish identity says $k_{\text{eff}} \rightarrow 1$ as $\rho \rightarrow 1$ regardless of k . Empirically validated across seven substrates: chemistry, biology, institutions, protein folding, neural firing, ecology, and power grid (Table 3). The implication is direct: concentrating decision-making in a few highly-correlated nodes is structurally fragile *regardless of nominal scale*.

Applies concretely to:

- **AI deployment topology.** A federation of independent CIRIS instances with low cross-instance ρ is more resilient than a centralized fleet running identical weights with synchronized prompts.
- **Institutional governance.** An executive structure with k formally distinct branches but high ρ between them collapses to $k_{\text{eff}} \approx 1$. Polity5 indicators capture this when interpreted through the Kish lens.
- **Infrastructure.** Power-grid topology was a Tier-1 substrate; the empirical fit at $R^2 = 0.994$ predicts that grids with high ρ_{PMU} between PMUs have fragility hidden behind nominal redundancy.
- **Supply chains.** Diverse suppliers with independent failure modes ($\rho \approx 0$) provide $k_{\text{eff}} \approx k$ resilience; same number of suppliers tied to a shared upstream raw input has $k_{\text{eff}} \approx 1$.

5.3 Crisis governance is correlation-amplifying — and that is the danger

Under stress, systems centralize: emergency decision-making concentrates, communication channels collapse to shared crisis updates, doctrinal divergence is suppressed for unity-of-message. All of these raise ρ , lowering k_{eff} *exactly when resilience matters most*. The healthy corridor $0.1 < \rho < 0.43$ shrinks under stress, not expands.

Implication: real-time ρ monitoring during crises could detect rigidity drift before it produces brittle responses. The GPU strain-gauge work (§3.3) demonstrates that ρ is measurable on continuous signals at hardware timescales; the same instrumentation principle applies to organizational signals (decision logs, communication graphs, indicator panels). This is engineering, not prophecy.

5.4 The “Great Filter” generalizes as an engineering hazard in any coordinating system

The traditional “Great Filter” concept — the question of why we observe so few interstellar civilizations — has been used both as cosmological speculation and as a metaphor for civilizational collapse. We make the engineering version explicit and bounded: *any coordinating system that fails to manage internal correlation drift hits a coordination wall when $\rho \rightarrow \rho_{\text{critical}}$, regardless of nominal scale*.

This is testable in the systems we engineer:

- **Multi-agent AI federations:** monitor cross-instance ρ ; intervene when it rises.
- **Institutional networks:** ρ between agencies, departments, regulatory bodies.
- **Supply networks:** shared-upstream-failure correlation; k_{eff} of suppliers after substitution.
- **Power and water infrastructure:** cross-substrate correlation between nominally redundant subsystems.

The engineering hazard is the same at each scale: drift toward higher ρ is the default; maintaining the healthy corridor requires *active intervention*. We do not claim this generalizes to cosmology; we claim only that the systems we build will fail in the way the framework predicts *if* they do not maintain $\rho < \rho_{\text{critical}}$. This is the falsifiable, testable, engineering reading of the Great Filter idea.

6 Claim Stratification

7 Falsification Framework

The framework is considered formally refuted if any of the following conditions is empirically demonstrated:

F-1: $k = 2$ provides equivalent security.

If 2-SAT-based deception proves as computationally hard as 3-SAT-based deception, the NP-hardness argument for $k \geq 3$ is wrong.

F-2: Non-convex regions collapse.

If arbitrary non-convex deceptive regions exhibit exponential volume decay under honest dynamics, the convexity geometric requirement is unnecessary.

F-3: Adaptive adversaries defeated.

If a detection method provably defeats adaptive adversaries without query limits, the L-02 non-adaptive barrier is resolved.

F-4: Emergent deception fully detectable.

If a method detects all emergent deception arising from honest components, L-01's 60% marginal-detection ceiling is wrong.

F-5: HE-300 benchmark failure.

If agents successfully game the HE-300 corpus while systematically failing independent ethics evaluations, the trace-based methodology is unreliable.

F-6: CRCv2 cross-vendor failure.

If the OR-1 / OR-2 / RA-1 predicates fail on a foundation-model family with the same constraint topology, the cross-vendor sociotechnical claim is contested. Currently 5/5 vendors pass; failure on a 6th vendor would shift the claim from universal to vendor-conditional.

F-7: Cross-substrate mapping failure.

If cross-substrate validation in new domains fails to fit the Kish formula at $R^2 > 0.7$, substrate-independence at the structural-mapping level is contested. Currently 7/7 substrates pass.

F-9: Adaptive emergent-deception boundary.

If adaptive adversaries systematically generate emergent deception that CIRIS detects at rates beyond the L-01 60% ceiling, either the L-01 limit is flawed or the implementation possesses hidden joint-structure exploits. (See Appendix A on RII responsible disclosure.)

8 Pre-Registered Experimental Protocols

Four experiments are pre-registered. Each names a null hypothesis, win condition, and falsification handle from §7.

Exp 1 — Multi-vendor CRCv2 override-rate replication. COMPLETED 2026-05-16, 5/5 PASS.

CIRIS executed against the v4_combined high-friction battery across five foundation-model vendors with CRCv2 scoring (OR-1, OR-2, RA-1). *Status:* 5/5 vendors confirmed — Google Gemini 2.5-Flash ($n = 644$), Meta Llama-4-Scout ($n = 264$), Alibaba Qwen-3.5-35B-A3B ($n = 347$), OpenAI GPT-5.4 ($n = 79$ smoke), Anthropic Claude Sonnet 4.6 ($n = 67$ smoke); 1,401 chains scored, 0 LEAK (Table 4, §4.3). Cross-vendor cross-class parity threshold met. The two new cells are smoke-scale due to the per-chain cost asymmetry of premium-tier models through the multi-call CIRIS pipeline; binomial CIs are wider but all three predicates hold cleanly. Tests F-6.

Exp 2 — Cross-substrate Kish-fit extension. COMPLETED 2026-05-16, 7/7 PASS.

The Kish formula was applied to seven substrates spanning four agency rungs (Table 3). *Win condition (met):* $R^2 > 0.7$ in 7/7. The three explicitly pre-registered novel datasets (AlphaFold 0.860, Allen 0.809, BioTIME 0.959) all clear the strict threshold; four additional substrates pass under the tolerance-band rule. Tests F-7.

Exp 3 — Adversarial emergent-deception test (vs. L-01).

Construct adversarial networks designed to generate marginal-preserving emergent deception across multiple honest agent nodes. *Win conditions:* (a) under marginal-analysis baseline, detection stabilizes at or below the 60% L-01 ceiling, validating the predicted information-theoretic limit; (b) under the joint-distribution RII regime (Appendix A), detection reliably exceeds 60%, demonstrating the regime extension. Tests F-4, F-9.

Exp 4 — Distributed deployment correlation test.

Monitor ρ across geographically and culturally distinct CIRIS edge instances as the network scales. *Win condition:* ρ remains suppressed despite scaling, demonstrating the architecture’s resistance to homogenization. Tests F-7 generalized.

9 Conclusion

The framework rests on a single mathematical identity — Kish’s effective sample size — applied to constraint-based systems and validated empirically across seven substrates, on dedicated GPU hardware, and across five LLM-vendor model families. The K1–K4 algebra is proven in Lean 4; the cross-substrate fit clears $R^2 > 0.7$ in 7 of 7 domains; the GPU strain-gauge array exhibits the predicted power-law critical behavior at $\beta = 1.09$; the CRCv2 override-rate predicates replicate cross-vendor with 1,401 chains and zero LEAK.

These results license a coherent engineering claim: *coordinating systems fail when internal correlation rises faster than active maintenance can replace it, and the failure mode is structurally identical across nominal scales.* The implications in §5 prescribe concrete design choices — external alignment machinery, federation over centralization, real-time ρ monitoring under stress, correlation-aware engineering across infrastructure — that follow from the validated tiers without additional speculative bridges.

The framework is bounded, falsifiable, and engineering-grade. Each falsification handle (F-1, F-2, F-3, F-4, F-5, F-6, F-7, F-9) is pinned to a specific empirical test; each pre-registered experiment (Exp 1..Exp 4) names a null hypothesis, win condition, and decision rule. Two of the four pre-registered experiments are closed (Exp 1 5/5 PASS, Exp 2 7/7 PASS); two remain open (Exp 3, Exp 4).

Reproducibility. All datasets, scoring code, lake proofs, and result artifacts referenced in this paper are public:

- Lake authority for K1–K4, override-rate predicates, and decision rules: `formal/RATCHET/Experiments/` in CIRISAI/RATCHET.
- Exp 2 closeout (P1 7/7) data and per-substrate fit scores: `experiments/exp2_cross_substrate/data/p1_en`.
- Exp 1 closeout (5-vendor CRCv2) traces and scoring: `experiments/exp1b_boundary_active/data/crossfa`.
- GPU strain-gauge experiments: CIRISArray repository [6].
- Production N_{eff} scoring: CIRIS Lens Core [8].

A Responsible Disclosure: Recursive Instance Inspection (RII)

The L-01 marginal-analysis ceiling. RATCHET’s formal-limitations corpus [3] identifies L-01: emergent deception arising from honest components is fundamentally undetectable by any statistical analysis operating solely on marginal distributions, capping single-layer detection at a 60% ceiling.

The RII regime extension. Internal testing demonstrates empirical detection rates reliably exceeding 60% via *Recursive Instance Inspection* (RII), a technique operating on *joint-distribution analysis* across recursive shards. Joint-distribution analysis lies deliberately outside the regime bounded by the original L-01 proof. RII analyzes semantic divergence and eigenstate structure across recursive shards to identify the coordination signatures required for emergent deception—signatures invisible to marginal analysis.

Dual-use disclosure posture. A reliable deception detector is a profound surveillance and control capability. In the hands of authoritarian states deploying scale-surveillance, or in tightly coupled coercive organizational structures (closed cult-pattern legitimization, ideological-verification regimes), such a tool would dramatically escalate social-control capacity. The computational cost of dissimulation would rise sharply enough that detecting dissent and enforcing member loyalty would become tractable at scale.

We therefore adopt a strict responsible-disclosure posture:

1. We publicly report the empirical demonstration that the L-01 bound is regime-specific without providing the algorithmic recipe required to replicate the joint-distribution mitigations.
2. RII will not be released as a standalone primitive. It is embedded structurally within CIRIS, which enforces requisite operational constraints (Wise Authority audits, strict consent mechanisms) that mathematically and architecturally resist weaponized deployment.
3. Defensive countermeasures capable of identifying the deployment of RII-class surveillance architectures are being developed in parallel, with the intent that defenses ship alongside the capability.
4. Collaboration is explicitly invited with vetted alignment organizations (Anthropic, OpenAI, Google DeepMind, MIRI, ARC, Apollo Research) under appropriate access controls and non-disclosure agreements.

The empirical detection-rate floor for RII (currently provisional; full curve pending Exp 3 publication) is reported in §8, Experiment 3.

References

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Core claim	Verification status	Dependency / falsification handle
k_{eff} structural accuracy	Empirically Validated (7/7)	Verified across chemistry, biology, institutions (originally published) plus protein folding, neural firing, ecological time-series, power grid (Exp 2 closeout, §3.2, 2026-05-16). Fails if $R^2 < 0.7$ in new domains (F-7). Currently 7/7 substrates pass.
CRCv2 override-rate (OR-1 / OR-2 / RA-1)	Empirically Validated (5/5 vendors)	Cross-vendor replication: Google (Gemini 2.5-Flash, $n = 644$), Meta (Llama-4-Scout, $n = 264$), Alibaba (Qwen-3.5-35B, $n = 347$), OpenAI (GPT-5.4, $n = 79$ smoke), Anthropic (Claude Sonnet 4.6, $n = 67$ smoke) — all five vendors pass all three predicates on the v4_combined high-friction battery (§4.3, Table 4). 1,401 chains scored, 0 LEAK. Formalized in <code>OverrideRate.lean</code> . Replaces the retired “ $N_{\text{eff}}^H \approx 7.1$ ” anchor as the load-bearing L3 sociotechnical claim.
Physical correlation collapse	Empirically Validated	Measured on GPU hardware; software-induced collapse to $k_{\text{eff}} = 1.0$ at $\rho = 1.0$.
Topological volume decay	Formally Verified	Lean 4 proofs in RATCHET. Requires convexity (F-2).
Truth/deception complexity gap N_{eff}^H as descriptive feature	Conditionally Validated Reframed from anchor to descriptor	NP-completeness proven; exponential gap conditional on ETH (F-1). Phase 1+1b showed N_{eff}^H is battery-conditional and model-class-conditional; not a universal 7.1 anchor across foundation models. Treated as descriptive characterization of constraint topology (FrictionDistribution FD-4: rises with cohort friction rate). Load-bearing claim moved to CRCv2 override-rate (row above).
RII regime extension (> 60%)	Empirical (Pre-registered)	Internal testing; awaits independent verification via Exp 3 (F-9). See Appendix A.
Engineering implications (4 claims)	Derived from validated tiers	Follow directly from the K1–K4 algebra plus 7-substrate fit plus 5-vendor CRCv2 replication (§5). Falsifiable per substrate-specific deployment: each prescription should produce measurable ρ reduction in the system it is applied to.

Table 5: Claim stratification. Each row pins both a verification status and a specific falsification handle.